

VOLUME 0

Master Guide and Table of Contents

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-000
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Revision B release position

REV B SUPERSEDES REV A
Rev A used a 7i97T-centered design, one 7i84U, incorrect resolver identifiers, retired evidence states, and an unsafe 7i84U TB1 polarity map. Rev B is rebuilt from the live 7i80HDT/7i44/7i49/two-7i84U authority and must be used in its place.

Area	Rev B consolidated position
Host	Mesa 7i80HDT over hm2_eth at 192.168.1.121
P1	7i44; physical channels 0/1 to 7i84U-A/B
P2	Plain 7i49; RES0/1/2 and AOUT0/1/2/3
P3	Empty; no bare 3.3 V GPIO field wiring
Field I/O	Two 7i84Us; 43 DI / 25 DO allocated, 21 DI / 7 DO reserve
7i84U TB1	1/2 VFIELDB, 3/4 VFIELDA, 5 VIN, 6/7/8 common
Firmware	Efinix resolver bitfile name/binary/provenance remain unverified
Evidence	Every physical channel prints its current authority_status

2. Volume map

Vol	Document	Title	Primary audience
0	MZK-LCNC-000	Master Guide and Table of Contents	All users
1	MZK-LCNC-001	Overview and Safety	Owner / safety reviewer
2	MZK-LCNC-002	Operation Manual	Qualified operator
3	MZK-LCNC-003	Mesa Conversion and Installation	Integrator
4	MZK-LCNC-004	Wiring Diagrams and I/O Reference	Electrician / integrator
5	MZK-LCNC-005	Maintenance and Troubleshooting	Maintenance
6	MZK-LCNC-006	Reference, Glossary, and Index	All users

3. Master table of contents

Volume	Sections
V0	Revision position; volume map; source hierarchy; pre-power dashboard; document control
V1	Machine overview; retained/replaced systems; safety authority; LOTO; E-stop; DC bus; Z brake; hazards
V2	Applicability; control station; power-up; homing; jogging; spindle; ATC; G-code; auxiliaries; shutdown; alarms
V3	Architecture; hardware; Ethernet; firmware; resolver interface; field power; HAL/INI; staged bring-up; rejected designs
V4	Five wiring figures; 7i49; 7i44 links; both 7i84U TB1 maps; all 96 digital I/O terminals; conflicts; status legend
V5	Maintenance intervals; resolver/analog/network health; motion/spindle/ATC troubleshooting; rollback; records
V6	OEM manual library; project file map; external sources; glossary; superseded claims; alphabetical index

4. Authority hierarchy

- Electrical channel authority: mesa/current_pin_authority.csv.
- Control logic and binding: linuxcnc/*.hal and mazak_vqc_20_40.ini must agree with the authority CSV.
- Narrative documents interpret the design but cannot override the CSV or HAL.
- These PDFs are release artifacts generated from the source commit printed on every page.
- OEM machine-side wiring remains governed by Mazak 41434WB and the verified as-built cabinet trace.

Authority status	Rows
COMMISSIONING_PENDING	51
DEFERRED	1
HOLD_CONFLICT	4
OPTIONAL_VERIFY	1
PROPOSED	39

Authority status	Rows
SPARE	35
UNBOUND	1

5. Pre-power deliverable dashboard

ID	Deliverable	Planning state
D1	As-built one-line + terminal plan	NOT DRAFTED
D2	Installed nameplate register	PARTIAL (checklists drafted; photos pending)
D3	Immutable Mesa firmware package	NOT ASSEMBLED (bitfile TBD)
D4	I/O checkout sheet	NOT DRAFTED
D5	Hardware E-stop risk assessment + schematic	PARTIAL (software chain drafted)
D6	Shared-bus precharge / discharge procedure	DRAFTED (measurements pending)
D7	Enable / fault / ready / Z-brake timing budget	TEMPLATE DRAFTED (measurements pending)
D8	Resolver phasing / commissioning procedure	PROCEDURE DRAFTED (measurements/traces pending)
D9	First-move plan	PROCEDURE DRAFTED (machine values/signatures pending)
D10	Velocity-mode tuning procedure + saved traces	PROCEDURE DRAFTED (traces pending)
D11	Physical travel / envelope survey	PARTIAL (Y done; X, Z pending)
D12	FR-SX speed / orient / gear state diagram	DIAGRAM DRAFTED (terminal/timing proof pending)
D13	ATC hazard analysis + dry-cycle fixture	NOT DRAFTED
D14	Network / real-time qualification + automatic response	PROCEDURE DRAFTED (logs pending)
D15	Restore / rollback package	TEMPLATE DRAFTED (evidence/photos/ledger pending)
D16	Signed hold points	NOT DRAFTED

NO SIGNED HOLD POINTS

D16 is not drafted. Control power, DC bus, amplifier enable, Z-brake release, spindle rotation, first motion, and automatic M6 remain blocked by their applicable D1-D15 prerequisites.

6. Document control

- Stable filenames are retained for Volumes 0-6 so links do not drift.
- The combined file is revisioned: Mazak_VQC2040_Complete_Manual_RevB.pdf.
- manifest.json records SHA-256 and page count for every released PDF.
- Regenerate with: `uv run scripts/build_manual_set.py`.
- Run `scripts/validate_authority.py` and `scripts/validate_control_logic.py` before release.

VOLUME 1

Overview and Safety

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-001
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

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This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Machine and retrofit scope

Machine: Mazak Vertical Quality Center VQC-20/40, serial 060231. Original control: Mazatrol M-2 / MELDAS. Shop supply is documented as 460 VAC three-phase into the retained machine transformer. The retrofit replaces the numerical-control and PLC interface functions while retaining the mechanical machine, Mitsubishi velocity-loop servo amplifiers, shared rectifier/capacitor bus, Tamagawa axis resolvers, FR-SX spindle drive, hydraulics, lubrication, coolant, and tool changer.

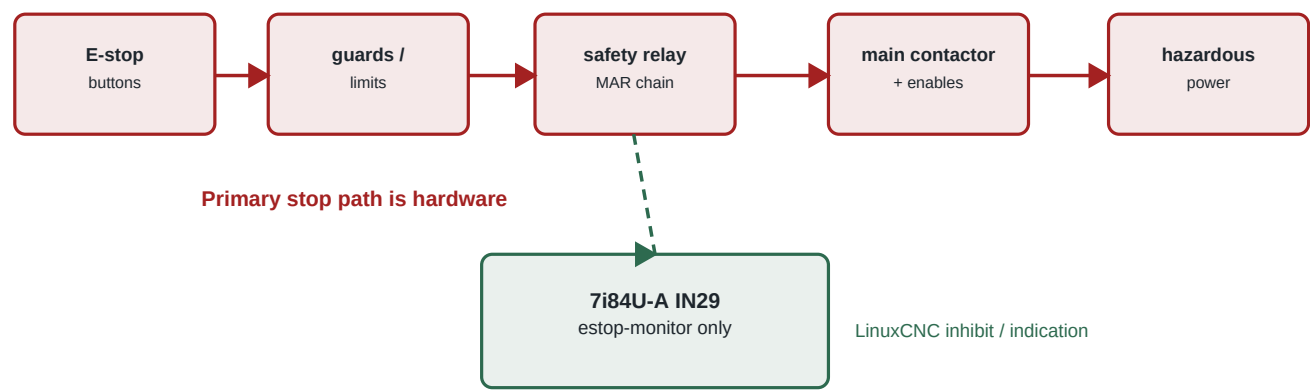
The retained equipment is obsolete and machine-specific. Adjacent-family Mitsubishi manuals are evidence for survey planning, not proof of the installed part or terminal behavior.

Function	Retained	Replaced / added
Motion power	MELDAS HD/TRA amplifiers, motors, tach loops, shared DC bus	LinuxCNC outer position loop and 7i49 analog commands
Feedback	Tamagawa TS2014N shaft resolvers	7i49 excitation and resolver-to-digital conversion
Spindle	Mitsubishi FR-SX, motor, orient detector, gearbox	7i49 AOUT3 plus 7i84U digital command/monitor
Field I/O	Switches, sensors, solenoids, relays	Two Mesa 7i84Us via 7i44
Safety	OEM MAR/hardwired chain intended to remain primary	LinuxCNC monitoring and supplemental inhibition only

2. Safety authority

LinuxCNC, hm2_eth, the 7i80HDT, 7i44, and 7i84U are not claimed as safety-rated equipment. The personnel-protection path must remove hazardous power through verified hardware without depending on the PC, Ethernet, FPGA, smart-serial, HAL, or the next servo cycle.

The checked-in estop-latch, watchdog, packet-error, drive-fault, and output-permit logic is a supplemental software inhibit. It cannot establish SIL, PL, stop category, guard integrity, or contactor redundancy.



The as-built hardware chain and fault-injection matrix are incomplete. No SIL/PL or stop-category claim is made.

Figure 1-1. Required authority split: hardware removes hazardous power; LinuxCNC monitors and inhibits.

3. Lockout, stored energy, and DC bus

- Open and lock the machine disconnect using the site's approved lockout/tagout procedure.
- Wait at least 10 minutes after power-off as required by the referenced Mitsubishi drive-family safety instructions.
- Do not trust elapsed time or a CHARGE lamp alone. Measure P(+) to N(-) with a properly rated meter and the exact installed-unit procedure.
- Proceed only below the service threshold specified by the identified rectifier/capacitor manual. The repository intentionally does not invent a universal voltage threshold.
- Repeat discharge characterization after rapid deceleration because regenerated energy can raise the bus above its at-rest condition.

- Support or crib the Z head before any brake, drive, or bus work.

SHARED BUS HAZARD

A single-axis fault does not isolate that amplifier from the common P/N bus. Bus discharge, regen, READY propagation, contactor behavior, and vertical-axis brake timing must be surveyed and measured as a group.

4. E-stop and guard-chain acceptance

- Draw the as-built chain with every button contact, guard, limit, safety relay, MAR relay, contactor, mirror contact, wire number, and removed load.
- Determine the achieved stopping category from hardware and measured stop behavior; do not assign Category 0 or 1 by assumption.
- Execute and sign the 18-case fault-injection matrix in docs/estop_safety_chain.md.
- Confirm manual reset is required after fault clearance and that communications recovery cannot automatically restore motion.
- Prove hard overtravel and torque removal without a software round trip through either 7i84U.

5. Z-axis gravity and brake hazard

The HAL uses an asymmetric placeholder sequence: S-ON request rises immediately, brake release follows after 0.100 s; on disable the brake command drops immediately and S-ON is retained for 0.100 s. These are unmeasured commissioning placeholders, not proof that the brake is mechanically holding before torque disappears.

Hardware E-stop, mains loss, internal amplifier fault, or loss of amp supply can bypass the software delay. The cabinet requires a proven brake-drop-before-torque-loss path, measured under the worst approved load and bus condition. Until then, treat every hard stop or power loss as capable of dropping the head.

NO PERSON UNDER Z

Do not stand under the spindle head or rely on the brake during commissioning. Use an engineered mechanical support whenever the Z holding chain is not proven.

6. Retrofit-specific hazard register

Hazard	Consequence	Required control / evidence
Reversed analog polarity	Immediate runaway	D8 direction proof, zero-voltage test, one-axis first-move plan
Wrong resolver winding/phasing	Wrong position, discontinuity, oscillation	Ohmmeter pairs, exact suffix datasheet, scope traces, independent scale proof
Dual resolver excitation	Damage or invalid feedback	Continuity/isolation proof; 7i49 sole source
Wrong 7i84U TB1 landing	Supply short or card damage	Pins 1/2 VFIELDB; 3/4 VFIELDA; 5 VIN; 6/7/8 common
Bare P3 field wiring	7i80HDT FPGA damage	P3 empty; probe on opto-isolated 7i84U-B IN15
100 VAC on 7i84U output	Card damage, shock	Interposing relays and load-specific suppression
Smart-serial/Ethernet loss	Delayed or missing control	Remote watchdog, HostMot2 watchdog, latched inhibit, measured fault tests
ATC state error	Tool drop, collision, pinch injury	D13 hazard analysis, dry-cycle fixture, mutual exclusion, no automatic retry
Unverified FR-SX interface	Unexpected rotation or failed orient	Exact drive manual, terminal trace, measured handshake/timing

7. Live-power hold points

Milestone	Required deliverables
Control power	D1, D2 partial, D3, D5, D15
DC bus	Control-power set plus D6
Amplifier S-ON	Above plus D4, D7, D14
Z-brake release	Above plus measured Z row of D7
Spindle rotation	Above plus D12
First axis motion	Above plus D8 and D9
First automatic M6	Above plus D10, D11, D13

VOLUME 2

Operation Manual

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-002
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

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This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

APPLICABILITY HOLD

This is a commissioning-stage operating framework, not authorization to run the machine. Normal operator procedures become valid only after the corresponding signed hold points, as-built wiring, and commissioned values are incorporated into a later revision.

1. Control station and current state

The planned interface is LinuxCNC Axis with an optional WHB04B-style USB pendant. Physical panel-button reuse is incomplete; the original Cycle Start input is deferred and the manual clamp input remains HAL-unbound. Do not assume an OEM button has retained its former function.

Both drive-output-permit and spindle-output-permit initialize FALSE. ATC DRY_RUN is 1. These deliberate holds are configuration controls and must be changed only by reviewed edits tied to signed commissioning steps.

2. Power-up framework

- Confirm the day's signed hold point authorizes the intended energy level; stop if it does not.
- Inspect the machine envelope, Z support state, tool/spindle condition, fluid leaks, cabinet condition, and all E-stop devices.
- Verify the dedicated LinuxCNC NIC is up at the recorded address and no unreviewed network or BIOS change occurred.
- Apply control power only. Confirm both commissioning holds are FALSE and all physical outputs remain safe.
- Review LinuxCNC messages, packet-error, watchdog, resolver error, drive-fault, E-stop monitor, and field-input states before any higher hold point.
- Use only the reviewed commissioning procedure for DC bus, S-ON, brake, spindle, or motion. Never bypass a linked output with ad-hoc halcmd writes.

3. Homing

The current INI is configured for switch-based homing in the order Z (HOME_SEQUENCE 0), X (1), Y (2). This is a skeleton configuration, not proof that Z is safe to move first. The resolver emulated index is not accepted for axis homing.

Before any homing attempt, verify raw/complement selection for all six limits and three home switches, prove the hardware overtravel path, measure travel and stop margin, and complete the Z-brake sequence if Z will move. Change homing order if the signed hazard review selects a safer sequence.

Axis	Configured limits (in)	Max vel / accel	Home sequence	Status
X / Joint 0	-39.4094 to 0.0394	1.0 / 5.0	1	PLACEHOLDER
Y / Joint 1	-30.0394 to 0.0394	1.0 / 5.0	2	PLACEHOLDER
Z / Joint 2	-18.1496 to 0.0394	1.0 / 5.0	0	PLACEHOLDER

4. Jogging and first motion

- Normal jogging is prohibited until D8 resolver proof, D9 first-move plan, D11 travel survey, D14 network qualification, and the relevant signed hold point are complete.
- The first move uses a normal closed-loop LinuxCNC jog, one physically isolated axis, at the recorded clamp and clearance. It does not use open-loop PID bias on an installed axis.
- Abort for wrong direction, unexplained zero-command motion, following error, resolver error, drive alarm, limit transition, packet error, watchdog, brake discrepancy, or loss of observer/instrument.

5. Spindle operation

The planned spindle path is 7i49 AOUT3 for analog speed and 7i84U-A OUT0/OUT1/OUT2 for FWD/REV/RUN. ORCM1 on OUT4 is a discrete orient command; it is not an analog position reference. Exact FR-SX model, terminals, polarities, analog mode, and READY handshake remain unverified.

Do not rotate the spindle until the spindle hold point is signed. Prove zero speed, at-speed, fault, run/direction exclusivity, analog scaling, gear confirmation, ORCM1-to-ORA1 timing, and cleanup after cancel/fault in both gear ranges.

6. Tool changer and magazine

Automatic M6 is blocked: [ATC] DRY_RUN remains 1, the Y soft-limit extension component is not implemented, the ATC hazard analysis/dry-cycle fixture is missing, and several solenoid identities/directions remain HOLD_CONFLICT.

A stalled or aborted ATC must not be jogged out under automatic power. Isolate energy, assess mechanical state, support any suspended tool, and use the approved manual recovery procedure after documenting the failure state.

ATC prerequisite	Current position
Spindle oriented	ORA1 planned on 7i84U-A IN4; terminal/timing unverified
Spindle zero speed	SZS planned on IN5; terminal/polarity unverified
Tool clamp / unclamp	PRS-9 IN15 and PRS-8 IN16; valve topology unverified
Magazine position	Five BCD inputs IN19-IN23, valid only with mag-in-pos IN28
Y/Z ATC zones	PRS-55 IN0 and PRS-66 IN1; fitment and dynamic-limit logic incomplete
Magazine direction	OUT13/OUT14 remain HOLD_CONFLICT pending SOL-8A/8B trace

7. G-code and tool data

- Use LinuxCNC G-code semantics, not Mazatrol conversational behavior, unless an explicit remap implements the function.
- Keep tool.tbl synchronized with verified physical pockets and the magazine BCD decode before any M6.
- Dry-run programs with spindle and drive outputs held off until coordinates, limits, units, work offsets, and remaps are independently checked.
- Parameter and program recovery from the M-2 is evidence for conversion/rollback; it is not automatically a LinuxCNC parameter import.

8. Coolant, air, hydraulics, and optional controls

Function	Authority	Restriction
Hydraulic/head-lube pump	7i84U-A OUT3	Proposed; prove pressure and contactor interface
Flood coolant pump	7i84U-A OUT11	Commissioning pending; interposing relay as required
Flood valve	7i84U-B OUT7	Proposed separate valve
Air / touch / tap blasts	7i84U-B OUT3/4/5	100 VAC legacy loads; RLY-5/6/7 required
Mist coolant	7i84U-A OUT12	HOLD_CONFLICT; do not energize
Magazine cover close	7i84U-B OUT8	Proposed single-coil function; trace before use
WHB04B pendant	USB / HAL	Optional; install only after base controls are proven

9. Shutdown and abnormal stop

- Use the reviewed normal stop sequence; wait for zero speed and confirmed stationary axes before machine-off.
- Confirm the Z brake is mechanically holding before torque is removed.
- Drop higher-energy hold points in the documented order; do not treat LinuxCNC exit as power isolation.
- After main power-off, retain lockout and measure the shared DC bus before service.
- After any E-stop, communications loss, drive alarm, resolver error, or ATC abort, require manual investigation and reset. Never auto-resume.

10. Operator alarm response

Indication	Immediate response
E-stop / hardware chain open	Keep clear; identify cause; do not reset until hardware state is known
Packet error or watchdog	Stop; preserve logs; inspect NIC/cable/bitfile; manual reset only after review
Resolver / following error	E-stop; isolate S-ON; preserve trace; do not retry by increasing limits
Drive fault	Isolate hazardous power; record amplifier indication; inspect shared READY propagation
Spindle orient timeout	Drop ORCM1/run permits; do not retry automatically; verify gear/zero-speed/ORAI
ATC abort	Isolate hydraulics/motion; document mechanism state; use approved manual recovery
Low air / hydraulic pressure	Inhibit ATC and clamp motion; correct supply/leak before reset

VOLUME 3

Mesa Conversion and Installation

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-003
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

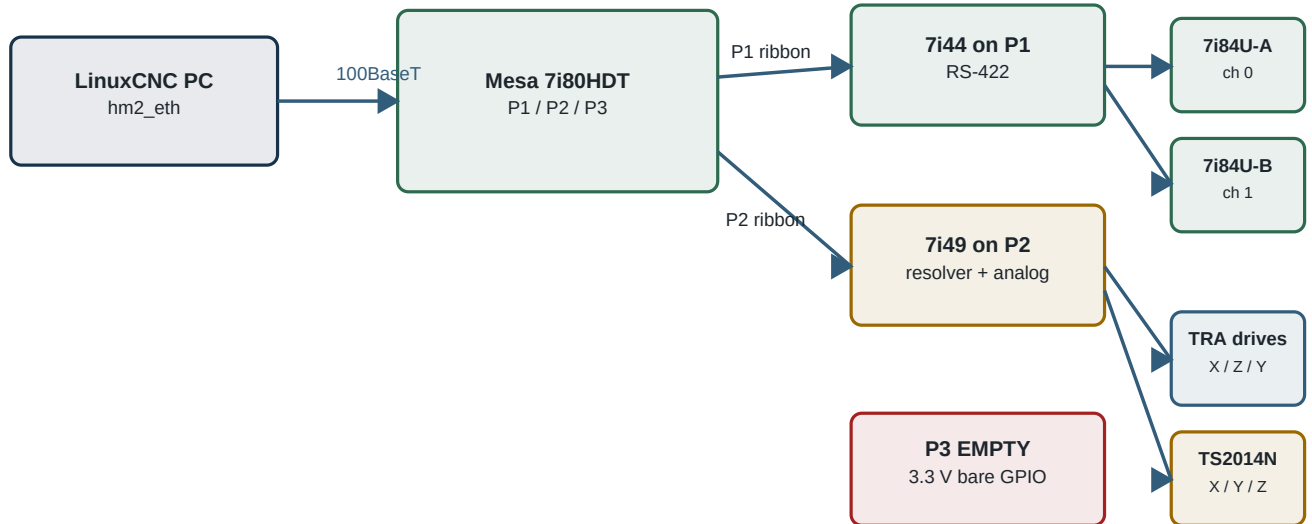
STATUS - NOT READY FOR POWER OR MOTION

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This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Selected architecture

The selected stack is a LinuxCNC PC connected by dedicated Ethernet to a Mesa 7i80HDT. P1 carries a 7i44 smart-serial interface to two 7i84U field-I/O cards. P2 carries a plain 7i49 for three resolver channels and four active analog outputs. P3 is intentionally empty.



Only the 7i49 carries motion feedback and analog commands. Field I/O is isolated through two 7i84Us. P3 has no field wiring.

Figure 3-1. Rev B control architecture. No 7i97T or 7i37TA is part of the selected stack.

Item	Role	Current qualification
LinuxCNC host	Trajectory, HAL, GUI	Debian 13 / LinuxCNC 2.9.10 / PREEMPT_RT target
Mesa 7i80HDT	100BaseT FPGA host, 3 x 50-pin connectors	Board/revision and loaded IDROM still to capture
Mesa 7i44 on P1	8-channel RS-422; channels 0/1 active	Physical link and exact HAL names unverified
Mesa 7i49 on P2	6 resolver + 6 bipolar analog channels	Plain board baseline; installed revision and signal levels unverified
7i84U-A ch 0	32 DI / 16 DO machine field I/O	Fully allocated; physical states unverified
7i84U-B ch 1	Limits, homes, probe, enables, relay loads	11 DI / 9 DO used; physical states unverified
P3	Future daughter-card position	Empty; bare 3.3 V GPIO must not see 24 V

2. Host and Ethernet setup

- Use a dedicated wired NIC directly connected to the 7i80HDT; no wireless or USB Ethernet.
- Board address is planned as 192.168.1.121; host control NIC is planned as 192.168.1.1/24 with no gateway. Verify the actual interface and MAC.
- Record NIC chipset, driver, firmware, negotiated 100 Mbit/s full duplex, IRQ, offload/coalescing state, and LinuxCNC startup diagnostics.
- Treat hardware-irq-coalesce-rx-usecs 0 as an A/B experiment; remove it if harmful on the installed NIC, especially affected Marvell hardware.
- Run at least 15 minutes idle latency, 4 hours worst-load latency, 2 hours with the real config and drives disabled, then cable-yank fault injection.
- Any packet-error event is investigated. packet-error-exceeded and watchdog.has_bit must latch motion inhibit and require manual recovery.

3. Firmware and IDROM gate

The working name 7i80hdt_7i44_ss_7i49d.bit is a placeholder only. Resolver configurations for the Efinix-based 7i80HDT are not proven by a stock-zip filename in this repository. Do not flash until Mesa/PCW provenance, binary, SHA-256, Efinity/source reference, recovery procedure, readhmid output, and HAL pin dump are committed as D3.

BITFILE HOLD

Never substitute a 7i80HD or 7i80HD-16 bitfile. The 7i80HDT uses a different FPGA and requires an HDT-specific build.

Expected loaded function	Planned config	Proof required
Resolvers	num_resolvers=3	IDROM + resolver.00-02 HAL pins
Analog PWM generators	num_pwmgens=4	pwmgen.00-.03 and measured AOUT mapping
Encoders	num_encoders=0	No spindle receiver/pins assigned
Smart serial	sserial_port_0=00xxxxxx	One port, channels 0/1, both remote serials
P3	IOPort only / unused	No field conductors; pin listing retained for future use

4. Resolver and analog motion path

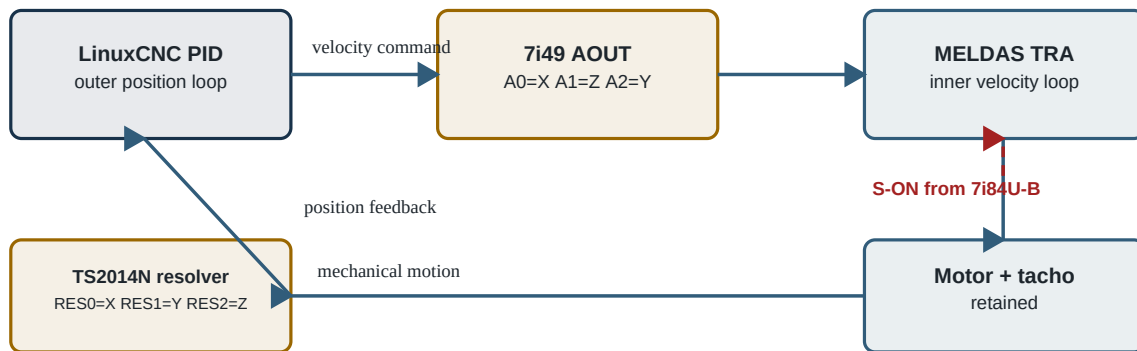


Figure 3-2. LinuxCNC closes the outer position loop; the retained TRA drive closes its inner velocity loop.

- Read and photograph the complete TS2014N suffix on X, Y, and Z. TS2014N141E26 values are comparison data only until suffixes are confirmed.
- Ohmmeter all winding pairs and compare with the exact datasheet. Do not trust old MELDAS signal names or colors.
- Use the 7i49 as the sole excitation source. The working 5 kHz selection is closest to a tentative 4.5 kHz variant but has no sourced tolerance acceptance.
- Scope RESDRV and SIN/COS at the card and resolver, at rest and through approved manual motion. Record amplitude, phase, offset, clipping, and noise.
- W2 affects only channels 3/4/5 and cannot correct X/Y/Z on channels 0/1/2.
- Derive signed resolver scale in machine units per resolver electrical revolution from screw travel and independent measurement. Do not leave 1.0 for motion.
- Calibrate volts-per-speed before tuning. OUTPUT_SCALE = 10 x measured velocity-per-volt; FF1 first, then P, with I/D only if evidence requires them.

Axis	Resolver	Analog output	Configured placeholders
X / J0	RES0	AOUT0 / pwmgen.00	res scale 1.0; output scale 1.0; gains zero
Z / J2	RES2	AOUT1 / pwmgen.01	res scale 1.0; output scale 1.0; gains zero
Y / J1	RES1	AOUT2 / pwmgen.02	res scale 1.0; output scale 1.0; gains zero
Spindle	Unassigned	AOUT3 / pwmgen.03	output scale 4000; command mode unverified

5. 7i84U field power and smart serial

Each 7i84U uses the conservative 5-28 VDC limit from the manual description/specification table. The project plans 24 VDC. Both VFIELDA and VFIELDB must be powered for their respective banks. Field power must not be mechanically switched in a way that violates the card's ramp requirement; use the documented output-enable/watchdog behavior and approved upstream design.

TB1 terminal	Function	Bank served
1 and 2	VFIELDB - both positive field supply terminals	TB2: IN16-31, OUT8-15
3 and 4	VFIELDA - both positive field supply terminals	TB3: IN0-15, OUT0-7
5	VIN logic power	W1 left links VIN to VFIELDB; W1 right needs separate VIN
6, 7, and 8	GND / VIN-VFIELD common	The only TB1 terminals used for 0 V return

TB1 POLARITY

Pins 2 and 4 are not returns. They are duplicate positive VFIELDB and VFIELDA terminals. Record each card's W1 position and never double-feed VIN.

6. Configuration files

File	Role
linuxcnc/mazak_vqc_20_40.ini	Machine/joint/spindle placeholders, ATC dry-run, timeouts, HAL load order
linuxcnc/mazak_vqc_20_40.hal	hm2_eth load, component scheduling, watchdog/E-stop/permit logic
linuxcnc/motion_7i80hdt.hal	7i49 resolver feedback, PID outer loops, pwmgen/analog outputs
linuxcnc/field_7i84u.hal	Both 7i84U pin bindings, limits, faults, enables, Z-brake sequence
linuxcnc/atc_orient.hal	ATC/orient component nets and output scheduling
linuxcnc/components/*.comp	Custom orient and ATC state machines
mesa/current_pin_authority.csv	Single physical channel authority; all wiring tables derive here

7. Staged bring-up

Stage	Required result
1	Complete applicable D1-D16 hold-point evidence; no live power before signatures.
2	Verify 7i80HDT communications, exact firmware/IDROM, HAL pins, NIC qualification, and fail-latched link-loss response.
3	Verify both 7i84U power maps, W1 positions, isolated 24 V buses, branch fusing, input modes, output load interfaces, and watchdogs.
4	With S-ON physically open, identify and energize resolver feedback only; scope all three channels and prove signed scale/direction manually.
5	With drives inhibited, verify AOUT0-AOUT3 zero voltage, polarity, mode, and measured volts-per-speed path.
6	Prove hardware E-stop, limits, drive faults, READY propagation, shared-bus stop/discharge, and Z-brake timing.
7	Execute the signed one-axis first-move plan, one axis physically enabled, at measured low authority. Do not assume Z first.
8	Tune each velocity-mode outer loop from measured scaling: FF1, P, then only evidence-driven I/D; save traces.
9	Commission spindle at low energy: analog scale, FWD/REV/RUN, zero speed, fault, gear, discrete orient, arrival/timeout cleanup.
10	Commission hydraulics and ATC output-by-output with no tool load, dry-cycle fixture, mutual exclusion, abort tests, and DRY_RUN held until acceptance.

8. Rejected and historical architectures

Item	Disposition
Mesa 7i97T	Retracted; not part of this analog-velocity/resolver stack
Mesa 7i80HD / 7i80HD-16	Different FPGA generation; never use its bitfile on a 7i80HDT
Mesa 7i37TA on P3	Retracted; duplicates field I/O and is not in the selected plan
Probe on P3 gpio.042	Retracted; unsafe 24 V on bare 3.3 V GPIO. Probe is 7i84U-B IN15
Analog orient reference on AOUT4	Retracted; ORCM1 is discrete and AOUT4 is spare
Quadrature axis encoders	Not selected; X/Y/Z feedback is resolver through 7i49

VOLUME 4

Wiring Diagrams and I/O Reference

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-004
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

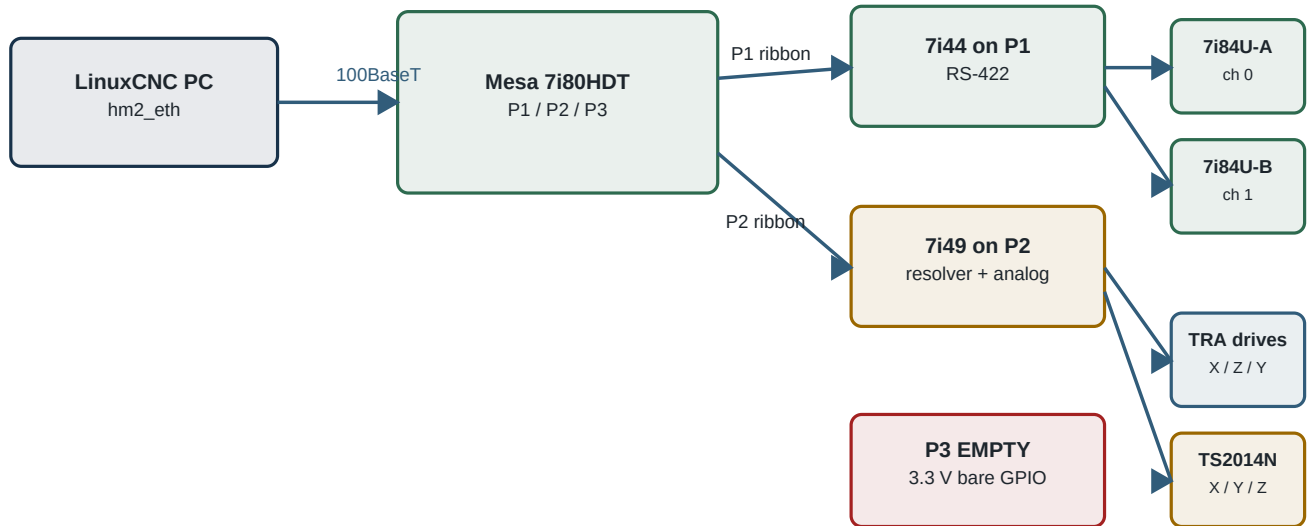
The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

WIRING AUTHORITY STATUS

Every table below is generated from mesa/current_pin_authority.csv. A row is not permission to land or energize a conductor unless its evidence state and applicable hold point allow it. HOLD_CONFLICT and UNBOUND rows are prohibited from field connection or actuation.

1. System architecture



Only the 7i49 carries motion feedback and analog commands. Field I/O is isolated through two 7i84Us. P3 has no field wiring.

Figure 4-1. Physical card topology and daughter-card ownership.

2. Servo command, enable, and feedback

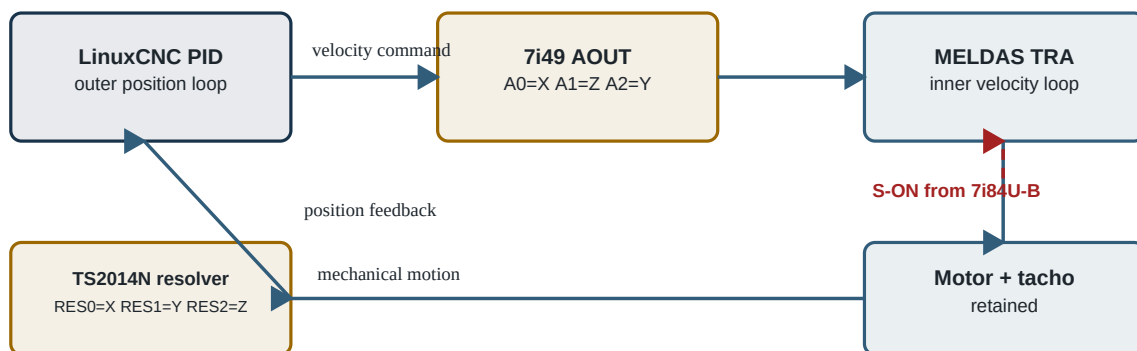
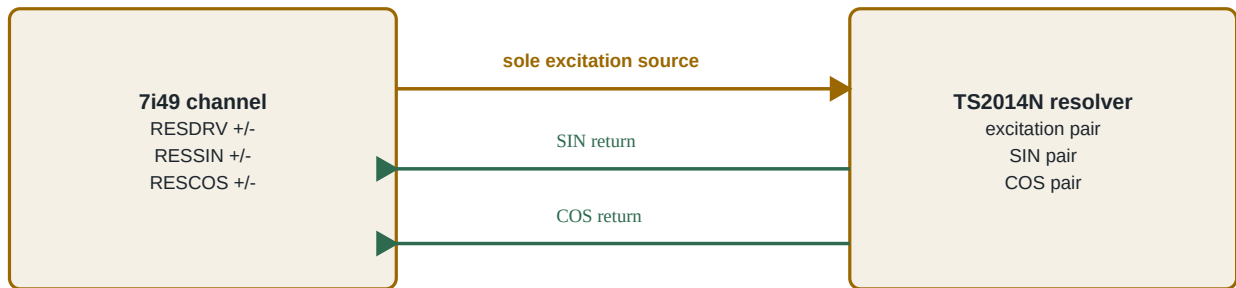


Figure 4-2. Velocity-mode nested loops and distinct 7i49/7i84U paths.

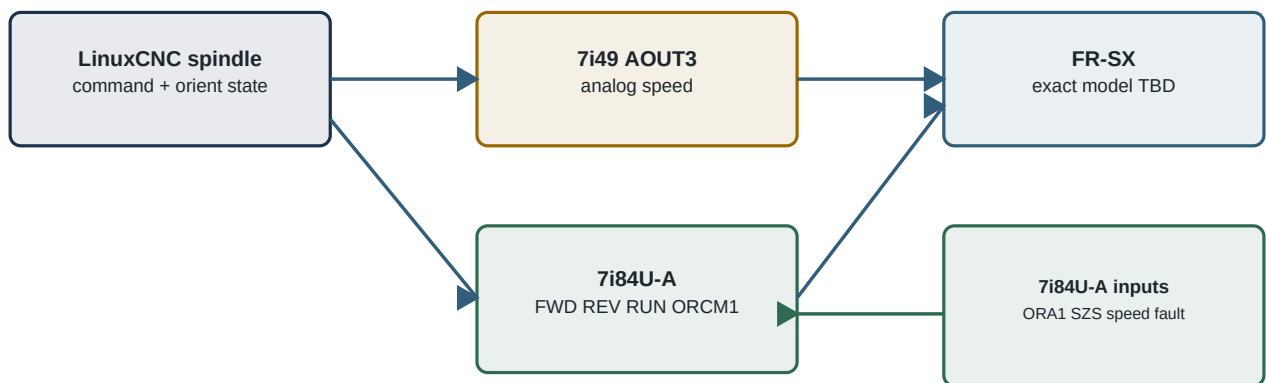
3. Resolver cable topology



Three individually shielded twisted pairs per axis; shields terminate at the 7i49 end only. Identify windings by resistance before power.

Figure 4-3. One 7i49 resolver channel; exact winding pairs are measured, not assumed.

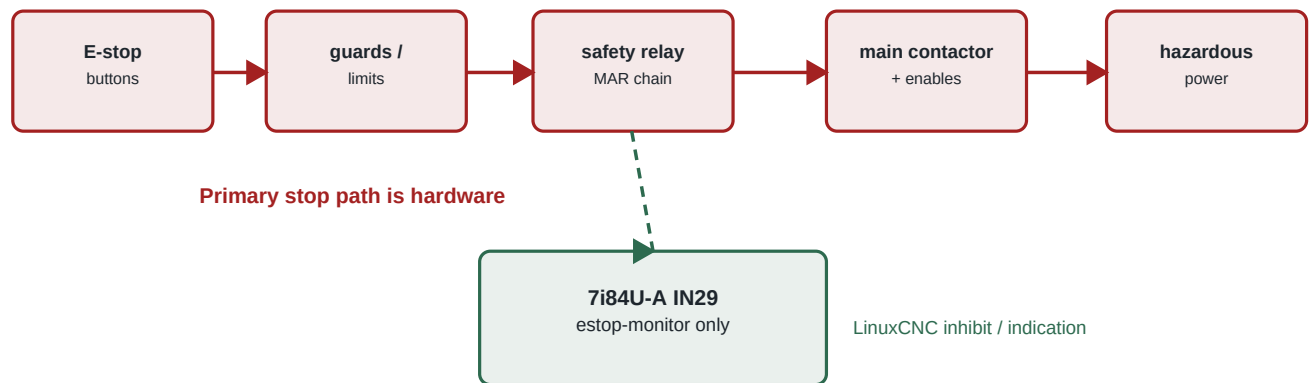
4. FR-SX spindle interface



Orient position is not an analog reference. ORCM1 is a discrete command; the machine sensor and drive parameters define the target.

Figure 4-4. Analog speed plus discrete run/orient model; exact FR-SX terminals remain unverified.

5. E-stop authority split



The as-built hardware chain and fault-injection matrix are incomplete. No SIL/PL or stop-category claim is made.

Figure 4-5. Hardware-primary stop path and software monitor tap.

6. 7i49 resolver and analog authority

Channel	Signal ID	HAL net	Field point / load	Status
RES0	X_RESOLVER	x-pos-fb	Tamagawa TS2014N X resolver	COMMISSIONING_PENDING
RES1	Y_RESOLVER	y-pos-fb	Tamagawa TS2014N Y resolver	COMMISSIONING_PENDING
RES2	Z_RESOLVER	z-pos-fb	Tamagawa TS2014N Z resolver	COMMISSIONING_PENDING
RES3	RES3_SPARE	none	7i49 spare resolver channel	SPARE
RES4	RES4_SPARE	none	7i49 spare resolver channel	SPARE
RES5	RES5_SPARE	none	7i49 spare resolver channel	SPARE
AOUT0	X_AXIS_CMD	x-vel-cmd	X servo analog velocity command to MELDAS TRA	COMMISSIONING_PENDING
AOUT1	Z_AXIS_CMD	z-vel-cmd	Z servo analog velocity command to MELDAS TRA	COMMISSIONING_PENDING
AOUT2	Y_AXIS_CMD	y-vel-cmd	Y servo analog velocity command to MELDAS TRA	COMMISSIONING_PENDING
AOUT3	SPINDLE_SPEED_CMD	spindle-speed-cmd	FR-SX V-IN speed reference	COMMISSIONING_PENDING
AOUT4	AOUT4_SPARE	none	Spare 7i49 analog output	SPARE
AOUT5	AOUT5_SPARE	none	Spare 7i49 analog output	SPARE

Axis analog order is X=AOUT0, Z=AOUT1, Y=AOUT2. Resolver order is X=RES0, Y=RES1, Z=RES2. These two orderings are deliberately different.

7. 7i44 smart-serial link authority

Channel	Signal ID	HAL net	Field point / load	Status
port0.TX+	SSERIAL_PORT0_TXA	none	7i84U-A RJ45 pin 2 RX+	PROPOSED
port0.TX-	SSERIAL_PORT0_TXB	none	7i84U-A RJ45 pin 1 RX-	PROPOSED
port0.RX+	SSERIAL_PORT0_RXA	none	7i84U-A RJ45 pin 6 TX+	PROPOSED
port0.RX-	SSERIAL_PORT0_RXB	none	7i84U-A RJ45 pin 3 TX-	PROPOSED
port0.GND	SSERIAL_PORT0_GND	none	7i84U-A RJ45 pin 4/5 ground	PROPOSED
port0.+5V	SSERIAL_PORT0_5V	none	7i84U-A RJ45 pin 7/8 +5V	PROPOSED
ports2-7	SSERIAL_PORTS_2_7_SPARE	none	Spare smart-serial channels	SPARE
port1.TX+	SSERIAL_PORT1_TXA	none	7i84U-B RJ45 pin 2 RX+	PROPOSED
port1.TX-	SSERIAL_PORT1_TXB	none	7i84U-B RJ45 pin 1 RX-	PROPOSED
port1.RX+	SSERIAL_PORT1_RXA	none	7i84U-B RJ45 pin 6 TX+	PROPOSED
port1.RX-	SSERIAL_PORT1_RXB	none	7i84U-B RJ45 pin 3 TX-	PROPOSED
port1.GND	SSERIAL_PORT1_GND	none	7i84U-B RJ45 pin 4/5 ground	PROPOSED
port1.+5V	SSERIAL_PORT1_5V	none	7i84U-B RJ45 pin 7/8 +5V	PROPOSED

8. 7i84U-A TB1 power map

Channel	Signal ID	HAL net	Field point / load	Status
TB1 pins 3/4 VFIELDA (+24 V field power, 5-28 VDC)	SEVENI84U_FIELD_A_24V	none	Field power bank A (TB3 I/O)	COMMISSIONING_PENDING
TB1 pins 1/2 VFIELDB (+24 V field power, 5-28 VDC)	SEVENI84U_FIELD_B_24V	none	Field power bank B (TB2 I/O)	COMMISSIONING_PENDING
TB1 pin 5 VIN (+24 V logic power)	SEVENI84U_VIN_24V	none	Field I/O logic power	COMMISSIONING_PENDING
TB1 pins 6/7/8 GND (VIN/VFIELD common)	SEVENI84U_GND	none	VIN/VFIELD common return	COMMISSIONING_PENDING

A-CARD TB1

Pins 1/2 are both VFIELDB positive; pins 3/4 are both VFIELDA positive; pin 5 is VIN; pins 6/7/8 are common. Use 5-28 VDC and record W1.

9. 7i84U-A inputs IN0-IN31

Channel	Signal ID	HAL net	Field point / load	Status
IN0	ATC_ZONE_Y	atc-y-zone	PRS-55 Y ATC zone	COMMISSIONING_PENDING
IN1	ATC_ZONE_Z	atc-z-zone	PRS-66 Z ATC zone	COMMISSIONING_PENDING
IN2	MAG_TOOL_AVAILABLE	mag-tool-avail	PHS-181 magazine tool available	COMMISSIONING_PENDING
IN3	SPINDLE_TOOL_AVAILABLE	spindle-tool-avail	PHS-182 spindle tool available	COMMISSIONING_PENDING
IN4	SPINDLE_ORIENT_ARRIVAL	spindle-oriented	FR-SX orient arrival (PLC X003 ORA1)	PROPOSED
IN5	SPINDLE_ZERO_SPEED	spindle-zero-speed	FR-SX zero-speed output (PLC X001 SZS.M)	PROPOSED
IN6	MAG_COVER_OPEN_CONF	mag-cover-open-conf	Magazine cover open reed switch (PLC X052 MGCORS)	PROPOSED
IN7	MAG_COVER_CLOSE_CONF	mag-cover-closed-conf	Magazine cover close reed switch (PLC X053 MGCCRS)	PROPOSED
IN8	THERMAL_ALARM_CHAIN	thermal-alarm	Motor thermal trip + main transformer overheat in series (PLC X073 THR.M + X07B ONT.M)	PROPOSED
IN9	MANUAL_TOOL_UNCLAMP_PB	manual-unclamp-pb	Manual tool unclamp switch at head (PLC X01A TUCFS.M)	PROPOSED
IN10	X_DRIVE_FAULT	x-drive-fault	X drive fault relay	COMMISSIONING_PENDING
IN11	Y_DRIVE_FAULT	y-drive-fault	Y drive fault relay	COMMISSIONING_PENDING
IN12	Z_DRIVE_FAULT	z-drive-fault	Z drive fault relay	COMMISSIONING_PENDING
IN13	SPINDLE_AT_SPEED	spindle-at-speed	FR-SX speed reach output	COMMISSIONING_PENDING
IN14	SPINDLE_FAULT	spindle-fault	FR-SX fault output	COMMISSIONING_PENDING
IN15	TOOL_CLAMP_CONF	tool-clamped	PRS-9 tool clamp confirm	COMMISSIONING_PENDING
IN16	TOOL_UNCLAMP_CONF	tool-unclamped	PRS-8 tool unclamp confirm	COMMISSIONING_PENDING
IN17	GEAR_HI_CONF	gear-hi-conf	PRS-10 gear high confirm	COMMISSIONING_PENDING
IN18	GEAR_LO_CONF	gear-lo-conf	PRS-12 gear low confirm	COMMISSIONING_PENDING
IN19	MAG_BCD_BIT0	mag-bcd-bit0	PRS-21 magazine BCD bit 0	COMMISSIONING_PENDING
IN20	MAG_BCD_BIT1	mag-bcd-bit1	PRS-22 magazine BCD bit 1	COMMISSIONING_PENDING
IN21	MAG_BCD_BIT2	mag-bcd-bit2	PRS-23 magazine BCD bit 2	COMMISSIONING_PENDING
IN22	MAG_BCD_BIT3	mag-bcd-bit3	PRS-24 magazine BCD bit 3	COMMISSIONING_PENDING
IN23	MAG_BCD_BIT4	mag-bcd-bit4	PRS-25 magazine BCD bit 4	COMMISSIONING_PENDING
IN24	DOOR_INTERLOCK	door-interlock	Door interlock switches series (LS-141 + LS-140 + PLC X01D ITMDSS.M all in one chain)	COMMISSIONING_PENDING
IN25	LUBE_OK	lube-ok	Two head-lube pressure switches (PLC X042 HLP2.M + X079 HLP.M) series-wired NC	PROPOSED
IN26	COOLANT_LOW	coolant-low	Coolant level switch	COMMISSIONING_PENDING
IN27	HYD_PRESS_OK	hydraulic-ok	Sanwa SPS-8T-PC-20 pressure switch	COMMISSIONING_PENDING
IN28	MAG_IN_POS	mag-in-pos	Magazine index in-position prox (PLC X00D MIPRS); BCD pot number valid only while TRUE	PROPOSED
IN29	ESTOP_MONITOR	estop-monitor	OEM MAR relay aux contact via interposing relay (Omron G2R-1-SND-DC24 or Phoenix PLC-RSC-24DC/21) driven from EHB bus	PROPOSED
IN30	MANUAL_TOOL_CLAMP_PB	manual-clamp-pb	Manual tool clamp switch at head (PLC X01B TCFS.M)	PROPOSED
IN31	SERVO_READY	servo-ready	Servo drives ready relay contact	COMMISSIONING_PENDING

10. 7i84U-A outputs OUT0-OUT15

Channel	Signal ID	HAL net	Field point / load	Status
OUT0	SPINDLE_FWD	spindle-fwd	FR-SX forward input	COMMISSIONING_PENDING
OUT1	SPINDLE_REV	spindle-rev	FR-SX reverse input	COMMISSIONING_PENDING
OUT2	SPINDLE_ENABLE	spindle-run-output	FR-SX enable input	COMMISSIONING_PENDING
OUT3	HYD_PUMP_ON	hyd-pump-on	Hydraulic + head-lube pump contactor (PLC Y096 HYD.M)	PROPOSED
OUT4	SPINDLE_ORIENT_CMD	spindle-orient-cmd	FR-SX orient command (PLC Y093 ORCM1.M)	PROPOSED
OUT5	SPINDLE_ORIENT_LOGEAR	orient-lo-gear	Low-gear orient assist (PLC Y094 CTL.M)	PROPOSED
OUT6	Z_BRAKE_REL_ENABLE	z-brake-release	N1J-L2-201 Z brake release	COMMISSIONING_PENDING
OUT7	GEAR_HI_SOL	gear-hi-sol	RLY-1 to SOL-13 Fujikoshi hydraulic valve	COMMISSIONING_PENDING
OUT8	GEAR_LO_SOL	gear-lo-sol	RLY-2 to SOL-12 Fujikoshi hydraulic valve	HOLD_CONFLICT
OUT9	TOOL_CLAMP_SOL	tool-clamp-sol	RLY-3 to SOL-10 Fujikoshi hydraulic valve	COMMISSIONING_PENDING
OUT10	TOOL_UNCLAMP_SOL	tool-unclamp-sol	RLY-4 to SOL-10 Fujikoshi hydraulic valve	COMMISSIONING_PENDING
OUT11	COOLANT_ON	flood-coolant	Coolant pump relay	COMMISSIONING_PENDING
OUT12	MIST_COOLANT	mist-coolant	Mist coolant relay (was documented as lube pump; verify at cabinet)	HOLD_CONFLICT
OUT13	MAG_CW_SOL	mag-cw-sol	ATC magazine CW rotation relay (SOL-8A per connector_crossref.md, unverified)	HOLD_CONFLICT
OUT14	MAG_CCW_SOL	mag-ccw-sol	ATC magazine CCW rotation relay (SOL-8B per connector_crossref.md, unverified)	HOLD_CONFLICT
OUT15	ALARM_OUT	alarm-out	Alarm light or horn	OPTIONAL_VERIFY

11. 7i84U-B TB1 power map

Channel	Signal ID	HAL net	Field point / load	Status
TB1 pins 3/4 VFIELDA (+24 V field power, 5-28 VDC)	SEVENI84UB_FIELD_A_24V	none	Field power bank A (TB3 I/O)	COMMISSIONING_PENDING
TB1 pins 1/2 VFIELDB (+24 V field power, 5-28 VDC)	SEVENI84UB_FIELD_B_24V	none	Field power bank B (TB2 I/O)	COMMISSIONING_PENDING
TB1 pin 5 VIN (+24 V logic power)	SEVENI84UB_VIN_24V	none	Field I/O logic power	COMMISSIONING_PENDING
TB1 pins 6/7/8 GND (VIN/VFIELD common)	SEVENI84UB_GND	none	VIN/VFIELD common return	COMMISSIONING_PENDING

B-CARD TB1

Pins 1/2 are both VFIELDB positive; pins 3/4 are both VFIELDA positive; pin 5 is VIN; pins 6/7/8 are common. Use 5-28 VDC and record W1.

12. 7i84U-B inputs IN0-IN31

Channel	Signal ID	HAL net	Field point / load	Status
IN0	X_LIMIT_PLUS	limit-x-plus	X positive overtravel limit (NC)	PROPOSED
IN1	X_LIMIT_MINUS	limit-x-minus	X negative overtravel limit (NC)	PROPOSED
IN2	Y_LIMIT_PLUS	limit-y-plus	Y positive overtravel limit (NC)	PROPOSED
IN3	Y_LIMIT_MINUS	limit-y-minus	Y negative overtravel limit (NC)	PROPOSED
IN4	Z_LIMIT_PLUS	limit-z-plus	Z positive overtravel limit (NC)	PROPOSED
IN5	Z_LIMIT_MINUS	limit-z-minus	Z negative overtravel limit (NC)	PROPOSED
IN6	X_HOME	home-x	X home switch (LS-42 assumed)	PROPOSED
IN7	Y_HOME	home-y	Y home switch (LS-52 assumed)	PROPOSED
IN8	Z_HOME	home-z	Z home switch (LS-62 confirmed TB-51)	PROPOSED
IN9	AIR_OK	air-ok	Machine air-pressure OK switch	PROPOSED
IN10	SEVENI84UB_IN10_SPARE	none	Spare input on 7i84U-B	SPARE
IN11	SEVENI84UB_IN11_SPARE	none	Spare input on 7i84U-B	SPARE
IN12	SEVENI84UB_IN12_SPARE	none	Spare input on 7i84U-B	SPARE
IN13	SEVENI84UB_IN13_SPARE	none	Spare input on 7i84U-B	SPARE
IN14	SEVENI84UB_IN14_SPARE	none	Spare input on 7i84U-B	SPARE
IN15	PROBE_SKIP1	probe-in	Renishaw MP-3 probe SKIP1 (PLC X03F SKIP1.M)	PROPOSED
IN16	SEVENI84UB_IN16_SPARE	none	Spare input on 7i84U-B	SPARE
IN17	SEVENI84UB_IN17_SPARE	none	Spare input on 7i84U-B	SPARE
IN18	SEVENI84UB_IN18_SPARE	none	Spare input on 7i84U-B	SPARE
IN19	SEVENI84UB_IN19_SPARE	none	Spare input on 7i84U-B	SPARE
IN20	SEVENI84UB_IN20_SPARE	none	Spare input on 7i84U-B	SPARE
IN21	SEVENI84UB_IN21_SPARE	none	Spare input on 7i84U-B	SPARE
IN22	SEVENI84UB_IN22_SPARE	none	Spare input on 7i84U-B	SPARE
IN23	SEVENI84UB_IN23_SPARE	none	Spare input on 7i84U-B	SPARE
IN24	SEVENI84UB_IN24_SPARE	none	Spare input on 7i84U-B	SPARE
IN25	SEVENI84UB_IN25_SPARE	none	Spare input on 7i84U-B	SPARE
IN26	SEVENI84UB_IN26_SPARE	none	Spare input on 7i84U-B	SPARE
IN27	SEVENI84UB_IN27_SPARE	none	Spare input on 7i84U-B	SPARE
IN28	SEVENI84UB_IN28_SPARE	none	Spare input on 7i84U-B	SPARE
IN29	SEVENI84UB_IN29_SPARE	none	Spare input on 7i84U-B	SPARE
IN30	SEVENI84UB_IN30_SPARE	none	Spare input on 7i84U-B	SPARE
IN31	SEVENI84UB_IN31_SPARE	none	Spare input on 7i84U-B	SPARE

13. 7i84U-B outputs OUT0-OUT15

Channel	Signal ID	HAL net	Field point / load	Status
OUT0	X_DRIVE_ENABLE	x-enable	X servo S-ON to MELDAS DK-427 (X-drive ENA input)	COMMISSIONING_PENDING
OUT1	Y_DRIVE_ENABLE	y-enable	Y servo S-ON to MELDAS DK-427 (Y-drive ENA input)	COMMISSIONING_PENDING
OUT2	Z_DRIVE_ENABLE	z-enable	Z servo S-ON to MELDAS DK-427 (Z-drive ENA input)	COMMISSIONING_PENDING
OUT3	AIR_BLAST	air-blast	SOL-62 via RLY-5 (interposing relay for 100VAC coil)	COMMISSIONING_PENDING
OUT4	TOUCH_SENSOR_BLAST	touch-sensor-blast	SOL-35 via RLY-6 (interposing relay for 100VAC coil)	COMMISSIONING_PENDING
OUT5	TAP_COOLANT_BLAST	tap-coolant-blast	SOL-61 via RLY-7 (interposing relay for 100VAC coil)	COMMISSIONING_PENDING
OUT6	ATC_BARRIER_SOL	atc-barrier	ATC barrier expand solenoid (PLC Y095 TCME.M)	PROPOSED
OUT7	FLOOD_VALVE	flood-valve	Flood coolant valve, separate from pump motor (PLC Y011 FCL)	PROPOSED
OUT8	MAG_COVER_CLOSE_SOL	mag-cover-close	Single energise-to-close magazine cover valve via interposing relay	PROPOSED
OUT9	SEVENI84UB_OUT9_SPARE	none	Spare output on 7i84U-B	SPARE
OUT10	SEVENI84UB_OUT10_SPARE	none	Spare output on 7i84U-B	SPARE
OUT11	SEVENI84UB_OUT11_SPARE	none	Spare output on 7i84U-B	SPARE
OUT12	SEVENI84UB_OUT12_SPARE	none	Spare output on 7i84U-B	SPARE
OUT13	SEVENI84UB_OUT13_SPARE	none	Spare output on 7i84U-B	SPARE
OUT14	SEVENI84UB_OUT14_SPARE	none	Spare output on 7i84U-B	SPARE
OUT15	SEVENI84UB_OUT15_SPARE	none	Spare output on 7i84U-B	SPARE

14. Unassigned, deferred, and P3 rows

Channel	Signal ID	HAL net	Field point / load	Status
none	SPINDLE_ENCODER	none	Machine-side A/B/Z spindle encoder if fitted	UNBOUND
none	CYCLE_START_PB	cycle-start-pb	Operator cycle start pushbutton	DEFERRED
TBD_FROM_IDROM	P3_GPIO_SPARE	none	Spare direct FPGA GPIO on P3 (all pins)	SPARE

15. Conflict and commissioning blocker register

Signal	Card/channel	Field point	Status	Required resolution
SPINDLE_ENCODER	none none	Machine-side A/B/Z spindle encoder if fitted	UNBOUND	Current target leaves P3 empty and requests num_encoders=0. Identify encoder model/electrical format and select a compatible receiver/daughter interface plus IDROM-proven pins before allocation.
GEAR_LO_SOL	7i84U-A OUT8	RLY-2 to SOL-12 Fujikoshi hydraulic valve	HOLD_CONFLICT	Conflict: connector_crossref.md omits SOL-12 and identifies SOL-13 as low; verify both gear-shift coils before wiring
MIST_COOLANT	7i84U-A OUT12	Mist coolant relay (was documented as lube pump; verify at cabinet)	HOLD_CONFLICT	CONFLICT: HAL wires iocontrol.0.coolant-mist to this output; CSV previously carried LUBE_ON. Trace physical load before commissioning; add row to wiring/authority_conflicts.md.
MAG_CW_SOL	7i84U-A OUT13	ATC magazine CW rotation relay (SOL-8A per connector_crossref.md, unverified)	HOLD_CONFLICT	CONFLICT #3 in wiring/authority_conflicts.md: SOL-8A/8B direction mapping unproven. Renamed from atc-fwd to match atc_orient.hal net. Verify direction under controlled commissioning.
MAG_CCW_SOL	7i84U-A OUT14	ATC magazine CCW rotation relay (SOL-8B per connector_crossref.md, unverified)	HOLD_CONFLICT	CONFLICT #3 in wiring/authority_conflicts.md: SOL-8A/8B direction mapping unproven. Renamed from atc-rev to match atc_orient.hal net. Verify direction under controlled commissioning.

16. Evidence-state legend

State	Meaning for wiring/use
PROPOSED	Paper design only; no physical verification
COMMISSIONING_PENDING	Defined but not physically accepted; treated conservatively like proposed
TRACED	De-energized conductor path and device action physically traced
ELECTRICALLY_VERIFIED	Normal/tripped voltage and load behavior measured
HAL_VERIFIED	Physical stimulus and unique HAL response captured
COMMISSIONED	Passed functional and applicable fault-injection acceptance
HOLD_CONFLICT	Contradictory identity/function; do not land or energize
UNBOUND	No hardware terminal/interface assigned
DEFERRED	Outside current scope
SPARE	Unassigned terminal; leave unlabeled

Maintenance and Troubleshooting

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-005
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Maintenance authority

Use Mazak 413m033 for retained machine maintenance, 41434WB for OEM electrical wiring, YM2V39L for original sequence meaning, and the exact Mitsubishi manual for each installed drive. This retrofit volume covers only added LinuxCNC/Mesa equipment and interfaces. It cannot replace model-specific service instructions.

2. Inspection schedule

Interval	Checks
Before each use	Leaks, loose tools, Z support/brake state, E-stops/guards, LinuxCNC faults, packet/watchdog state, abnormal noise
Weekly	Cabinet filters/seals, fan operation, terminal discoloration, relay indicators, 24 V rails, cable abrasion, coolant intrusion
Monthly	PE bonds, shield drains, connector retention, 7i44/7i84U links, output relays/suppression, log/backup completeness
Quarterly	E-stop/guard functional checks per approved matrix subset, brake holding trend, bus discharge trend, resolver/noise survey comparison
Annually	Full safety validation, torque/terminal inspection per OEM specs, resolver amplitude/phase baseline, analog zero, NIC/latency requalification after updates
After any change	Firmware hash/IDROM, HAL pins, authority validator, control validator, affected hold point and fault-injection tests

3. Safe diagnostic order

- Preserve the first fault: messages, indicator states, HAL values, scope trace, and physical relay state before reset.
- Separate control indication from physical energy. A false HAL pin does not prove a contactor, brake, or bus is safe.
- Isolate and measure the shared P/N bus before opening drive wiring.
- Use the authority CSV to identify the exact card/channel and status; stop if HOLD_CONFLICT or UNBOUND.
- Test from field device toward HAL for inputs, and from reviewed request logic toward load for outputs. Never bypass interlocks in the active machine config.
- After repair, repeat every safety/functional acceptance affected by the changed component or conductor.

4. Motion and resolver troubleshooting

Symptom	Likely areas	Required action
Position noisy at rest	Resolver pair/shield, dual excitation, low return, switching noise	S-ON off; scope differential RESDRV/SIN/COS; compare baseline; operate nearby relays one at a time
Wrong feedback direction	Signed resolver scale or phasing	Return to D8 manual direction proof; do not negate PID gain
Axis moves at zero command	Analog offset, enable/gate error, drive null	E-stop; open S-ON; measure AOOUT and pid/pwmgen values; require <=50 mV before retry
Following error at constant speed	OUTPUT_SCALE/FF1 mismatch, drive gain, resolver scale	Return to volts-per-speed calibration; do not simply raise ERROR
Oscillation/whine	P too high, TRA inner loop, resolver noise	Rollback to last logged settings; inspect drive loop and feedback before retuning
Single drive fault drops group	Shared READY chain behavior	Expected design premise but machine-specific propagation must be logged; inspect all axes and bus
Z drops on stop	Brake sequencing/holding failure	Support Z, isolate power, quarantine motion; investigate hardware timing and brake capacity

5. Ethernet, FPGA, and smart-serial troubleshooting

Symptom	Checks	Do not
7i80HDT unreachable	Dedicated NIC address/link, cable, board power/IP, routing	Do not flash firmware as a connectivity test
packet-error-level rises	NIC driver/IRQ/load, cable, coalescing/offload A/B record, servo latency	Do not increase limits until the cause is understood
watchdog.has_bit	HostMot2 scheduling, thread overrun, Ethernet/FPGA loss	Do not auto-reset into motion
7i84U missing	7i44 channel cable/pinout, remote power/W1, config mask, IDROM	Do not assume device tag or sserial mask
Remote outputs drop	Remote watchdog time, link, field bank voltage, VIN, output fault	Do not disable the watchdog
HAL pins differ	Loaded bitfile/readhmid/config string	Do not rename around an unproven firmware mismatch

6. Spindle and orient troubleshooting

Symptom	Checks	Safe response
No spindle rotation	Static/dynamic permits, READY, fault, FWD/REV/RUN, AOUT3, gear confirm	Keep low energy; verify exact FR-SX terminal states
Wrong direction	FWD/REV exclusivity and analog sign/mode	Drop all spindle permits; correct one documented sign stage
No orient arrival	Zero speed, gear, ORCM1, detector/parameter mode, ORA1 polarity	Allow component timeout; no automatic retry
Orient cancel restarts spindle	Run bit remained asserted; adjacent-family behavior	Treat as hazardous until exact FR-SX command order is proven
At-speed unreliable	IN13 source, polarity, threshold, debounce	Do not bypass ATC spindle-state interlock

7. ATC and field-I/O troubleshooting

Symptom	Checks	Safe response
Magazine position wrong	IN19-IN23 BCD, IN28 validity, pocket/tool table	Stop indexing; compare each bit at a known pocket
Clamp state disagrees	PRS-8/9, pressure, SOL-10 topology, OUT9/10	Support tool; isolate hydraulics; resolve valve mapping
Gear does not confirm	Zero speed, OUT7/OUT8, PRS-10/12, coil identity	Do not spin; HOLD_CONFLICT remains until traced
Output does nothing	Field bank/VIN, W1, watchdog, interposing relay, coil voltage/current	Measure at each boundary; never apply 100 VAC to 7i84U
Unexpected adjacent input	Commoning, sensor wiring, input mode, induced noise	Do not promote to HAL_VERIFIED; trace and repeat
ATC stalls	Current state, prerequisites, timeout/alarm, hydraulic pressure	No recovery jog; isolate and use approved manual recovery

8. Backup, restore, and rollback

- Maintain an immutable original-control photo/wire ledger and M-2 parameter/ladder backups before removing conductors.
- For every released retrofit state, archive the git commit, authority CSV, INI/HAL, custom components, tool table, bitfile binary/hash/source, readhmid, HAL pin dump, host package/kernel/NIC record, and signed test evidence.
- Tag a known-safe baseline only after its hold-point evidence is complete. A git tag without physical acceptance is not a safe machine baseline.
- After a fault during commissioning, return both output holds FALSE, physically isolate S-ON, restore the logged low-authority configuration, and preserve all evidence before retry.
- Reverting to the original M-2 requires the removed-conductor ledger and independent continuity check; never reconnect from memory or an old Rev A PDF.

9. Spares and obsolescence

Mitsubishi TRA/HD and FR-SX hardware is obsolete. Record installed suffixes before purchasing used spares. Store repaired/used units with test reports and retain parameter/trim positions. Mesa spares must match board revision and verified firmware package; a replacement 7i80HDT must not be flashed from an assumed filename.

10. Required maintenance records

Record	Location
I/O checkout and state measurements	wiring/io_checkout_sheet.md + docs/commissioning_logs
Resolver scope/phasing/scale	docs/commissioning_logs/resolver_traces
Servo tuning and faults	docs/commissioning_logs/tuning and per-axis logs
Bus discharge/regen	docs/commissioning_logs/dc_bus_discharge_*
E-stop fault matrix	docs/commissioning_logs/estop_validation_*
Network/latency/cable-yank	docs/commissioning_logs/latency_*
Firmware package	mesa/firmware with SHA256SUMS/readhmid/HAL pins
Configuration release	Git commit/tag plus manual-set manifest.json

VOLUME 6

Reference, Glossary, and Index

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-006
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Original Mazak manual library

Publication	Title	Use
413S038	Operating Manual for VQC-20/40 and 20/50	Primary retained-machine operation reference
M00S075	Operating Manual for CAM M-2	Original conversational control reference
413M033	Maintenance Manual for VQC-20/40 and 20/50	Primary retained mechanical/electrical maintenance
41434WB	Electrical Circuit Diagram for VQC-20/40/50	Primary OEM machine-side wiring source
YM2V39L	PLC Ladder Diagrams	Original sequence and signal meaning
PAREXM210E	Parameter List and Explanation for M-2	M-2 parameter definitions
M00P111	Programming Manual for CAM M-2 EIA/ISO	Original M-2 G-code reference
M00P135	Programming Manual for M-2E	Reference only; machine is M-2
M00P018	Programming Manual for Mazatrol CAM 3D	Option reference; fitment unverified
M00T002	Tooling Manual for Mazak Machining Center	Toolholder and pull-stud reference
413LE02A000	Parts List for VQC 20/40 and 20/50	Mechanical parts sourcing

The 41434WB drawing title-block history includes M-1-era sheets although serial 060231 carries an M-2 control. Use the drawings for retained machine wiring and verify the installed NC/control revision separately.

2. Key project files

File	Authority / purpose
mesa/current_pin_authority.csv	Physical channel authority and evidence state
docs/architecture_decision.md	Selected 7i80HDT/7i44/7i49/two-7i84U architecture
linuxcnc/*.hal and *.ini	Active logic skeleton and configuration placeholders
docs/pre_power_deliverables.md	D1-D16 evidence and hold-point charter
docs/estop_safety_chain.md	Hardware safety requirements and 18-case fault matrix
docs/dc_bus_stop_fault.md	Shared-bus survey/discharge/regen/fault plan
docs/resolver_commissioning.md	Per-axis winding, phasing, scale, and noise proof
docs/servo_commissioning.md	Velocity-mode calibration and FF1-first tuning
docs/hm2_eth_nic_validation.md	Debian 13/NIC/latency/packet/watchdog qualification
docs/frsx_orient_model.md	Discrete-orient model and unresolved FR-SX identity
docs/first_move_plan.md	Signed one-axis first-motion procedure
scripts/validate_authority.py	CSV/HAL/capacity/legend/TB1 regression checks
scripts/validate_control_logic.py	Static HAL/INI safety invariants
scripts/build_manual_set.py	Reproducible Rev B PDF generator

3. Primary external references

Source	Use
Mesa 7i80HDT product page	Host capability: 100BaseT, 72 I/O, three 50-pin connectors, 5 V tolerant
Mesa 7i44 manual	Eight RS-422 channels and physical serial pinout
Mesa 7i49 manual	Six resolver channels, six bipolar analog outputs, cabling, excitation, W2
Mesa 7i84U manual	32 DI/16 DO, TB1/TB2/TB3, 5-28 VDC specification, watchdog/update facts
LinuxCNC hostmot2(9)	Resolver/PWM/sserial/watchdog module semantics
LinuxCNC hm2_eth(9)	Dedicated NIC, packet error, timeout, firewall, coalescing guidance
LinuxCNC pid(9)	FF terms, output engineering units, clamps and windup
Mitsubishi TRA/MELDAS drive manuals	Shared bus, READY, regen, discharge, amplifier setup
Exact Tamagawa suffix datasheets	Required per axis; not yet captured
Exact installed FR-SX manual	Required; model/terminal layout not yet captured

4. Glossary

Term	Definition
AOUT	7i49 bipolar analog output used as a velocity command
Authority	The controlled pin-to-signal record in current_pin_authority.csv
COMMISSIONING_PENDING	Defined design row without completed physical acceptance
D1-D16	Pre-power evidence deliverables and signed hold-point prerequisites
FF1	PID velocity feed-forward term after engineering-unit output scaling
FR-SX	Retained Mitsubishi spindle drive family; exact installed model remains unverified
HAL	LinuxCNC Hardware Abstraction Layer
HOLD_CONFLICT	Contradictory field identity/function; connection or actuation prohibited
hm2_eth	LinuxCNC Ethernet driver for HostMot2 boards
IDROM	Firmware module and pin-routing identity data read from the FPGA image
MAR	OEM relay/chain designation associated with machine-ready/E-stop behavior; trace as-built
ORCM1	Discrete spindle orient command, OEM wire Y093
ORA1	Spindle orient-arrival indication, OEM wire X003
P/N bus	Shared DC bus from retained rectifier/capacitor stack
P3	Unused 7i80HDT 50-pin connector; bare 3.3 V GPIO, no field wiring
PROPOSED	Paper design only; no physical verification
RESDRV	7i49 differential resolver excitation pair
RESSIN/RESCOS	7i49 differential resolver return pairs
S-ON	Servo amplifier enable input
sserial	Mesa smart-serial link; two 7i84Us on 7i44 channels 0/1
TRA	Retained Mitsubishi analog velocity-loop servo amplifier family
TS2014N	Tamagawa shaft-resolver family; exact suffix required per axis
VFIELDA	7i84U field supply for TB3 I/O; TB1 pins 3/4 both positive
VFIELDDB	7i84U field supply for TB2 I/O; TB1 pins 1/2 both positive
VIN	7i84U logic supply on TB1 pin 5; W1 can link it to VFIELDDB
W1	7i84U jumper selecting linked or separate VIN feed
W2	7i49 half-drive option affecting channels 3/4/5 only
Watchdog	Hardware/remote timeout that must fail outputs off and latch recovery

5. Superseded claim quarantine

Retracted / unsafe claim	Current disposition
7i97T is the primary controller	Retracted; 7i80HDT host with 7i49/7i44 daughters
7i80HD-16 is interchangeable	Contradicted; never flash HD/HD-16 images on HDT
7i37TA is on P3	Retracted; P3 is empty
Probe uses P3 gpio.042	Retracted; 7i84U-B TB3 IN15
Axis feedback is quadrature encoder	Contradicted; resolver through 7i49
AOUT4 commands orient angle	Contradicted; ORCM1 is discrete and AOUT4 spare
W2 can correct X/Y/Z amplitude	Contradicted; W2 affects channels 3/4/5 only
5 kHz is within a published TS2014N tolerance	Unverifiable; exact suffix and scoped measurement required
RESOLVER_SCALE=1.0 is commissioned	Unsafe placeholder; measure signed machine units/electrical rev
OUTPUT_SCALE=MAX_VELOCITY is automatically correct	Only true if measured drive gain makes 10 V equal that velocity
Use fixed P gain 10-30	Rejected; no universal gain; select from measured response
Use 250 ms Z off-delay	Rejected without brake/relay/drive measurements
Probe <=15 IPM is proven safe	Rejected without end-to-end latency, stopping, and repeatability measurements
Disable all NIC RX/TX checksum offloads	Not universal; A/B test supported settings on the installed NIC
7i84U pins 2/4 are returns	Dangerous error; both are positive field-supply duplicates

6. Alphabetical index

Entry	Reference
7i44	V3 s1/s5; V4 s7
7i49	V3 s4; V4 s2/s3/s6
7i80HDT	V3 s1-s3; V4 s1
7i84U-A	V4 s8-s10
7i84U-B	V4 s11-s13
ATC	V2 s6; V4 s9-s13; V5 s7
Analog command scaling	V3 s4; V5 s4
Architecture	V0 s1; V3 s1; V4 s1
Authority hierarchy	V0 s4; V4 s16
Bitfile	V3 s3; V5 s5
Brake, Z axis	V1 s5; V4 s10; V5 s4
DC bus	V1 s3; V5 s3
E-stop	V1 s2-s4; V4 s5
Ethernet / hm2_eth	V3 s2; V5 s5
Evidence states	V0 s4; V4 s16
FR-SX	V2 s5; V4 s4; V5 s6
Field power	V3 s5; V4 s8/s11
Firmware	V3 s3; V5 s5
First motion	V2 s4; V3 s7
Grounding / shielding	V1 s6; V3 s4; V4 s3
HAL files	V3 s6
Hold points	V0 s5; V1 s7
Homing	V2 s3
I/O map	V4 s6-s14
Interposing relays	V1 s6; V2 s8; V4 s10/s13
Maintenance	V5 s1-s10
Manual library	V6 s1
Network faults	V1 s6; V5 s5
P3	V3 s1/s8; V4 s14
Probe	V2 s4; V3 s8; V4 s12
Resolver	V3 s4; V4 s2/s3/s6; V5 s4
Rollback	V2 s9; V5 s8
Safety	V1 s1-s7
Servo tuning	V3 s4/s7; V5 s4
Shared bus	V1 s3; V5 s4
Smart serial	V3 s5; V4 s7; V5 s5
Spindle orient	V2 s5; V4 s4; V5 s6
Status codes	V4 s16
TB1 power	V3 s5; V4 s8/s11
Tool changer	V2 s6; V5 s7
Troubleshooting	V5 s3-s7
Watchdog	V1 s6; V3 s2/s5; V5 s5